



BCFireCube Codebook

Williams, Zane¹

`zane.williams@ubc.ca`

Baylis, Patrick¹

`patrick.baylis@ubc.ca`

¹Vancouver School of Economics, The University of British Columbia, Canada

Abstract

We introduce the BCFireCube, a harmonized datacube compiled to support wildfire research in British Columbia. Previous works have produced vital spatial data describing British Columbia's wildfires, ecology, geography, and social development. However, inconsistent data formatting, limited resource availability, and poor data interoperability hinder research. The BCFireCube addresses this barrier by modifying and harmonizing spatial data produced by the Government of British Columbia, Natural Resources Canada, and Statistics Canada. We provide spatial, ecological, and social data about wildfires from 1917 to 2025, in addition to 21 congruent raster layers and 3 vector files which contextualize British Columbia's wildfire regime. This harmonized data is available on the Borealis, the Canadian Dataverse Repository. The BCFireCube has a wide range of potential applications in wildfire science, wildfire economics, political ecology, and environmental risk modelling.

Contents

Directory Tree	3
Wildfire Data	4
Point and Polygon Data	4
Fire Points	8
Fire Polygons	9
Point Data	10
Polygon Data	13
Raster Data	16
Dwellings	16
Land Cover	17
Population	19
Digital Elevation Model	20
Slope	21
Wildland-Urban Interface Buffer	22
Biogeoclimatic Zone and Subzone	23
Natural Disturbance Type	30
PSTA Fire Density	31
PSTA Head Fire Intensity	33
PSTA Human Fire Density	35
PSTA Lightning Fire Density	37
Forest Total Biomass	39
Forest Age	40
Forest Mean Height	41
Forest Height Standard Deviation	42
Fuel Types	43
Forest Cover Ownership	45
Differenced Normalized Burn Ratio	47
NBAC Burn Year	48
Salvage Logging	49
Vector Data	50
Terrestrial Extent	50
Fire Centres and Zones	51
Watershed Groups	52

Directory Tree

```
BCFireCube
├── bcfire
│   ├── 2026_bcfire_bothdata.csv
│   ├── 2026_bcfire_firepoints.gpkg
│   ├── 2026_bcfire_firepolygons.gpkg
│   ├── 2026_bcfire_pointdata.csv
│   └── 2026_bcfire_polygondata.csv
├── bcrast
│   ├── dwellings
│   │   ├── 2006_bcrast_dwellings.nc
│   │   ├── 2011_bcrast_dwellings.nc
│   │   ├── 2016_bcrast_dwellings.nc
│   │   └── 2021_bcrast_dwellings.nc
│   ├── landcover
│   │   ├── 2010_bcrast_landcover.csv
│   │   ├── 2010_bcrast_landcover.nc
│   │   ├── 2015_bcrast_landcover.csv
│   │   ├── 2015_bcrast_landcover.nc
│   │   ├── 2020_bcrast_landcover.csv
│   │   └── 2020_bcrast_landcover.nc
│   ├── population
│   │   ├── 2006_bcrast_population.nc
│   │   ├── 2011_bcrast_population.nc
│   │   ├── 2016_bcrast_population.nc
│   │   └── 2021_bcrast_population.nc
│   ├── 2014_bcrast_dem.nc
│   ├── 2014_bcrast_slope.nc
│   ├── 2020_bcrast_wui.nc
│   ├── 2021_bcrast_bec.csv
│   ├── 2021_bcrast_bec.nc
│   ├── 2021_bcrast_ndt.csv
│   ├── 2021_bcrast_ndt.nc
│   ├── 2021_bcrast_pstafire.csv
│   ├── 2021_bcrast_pstafire.nc
│   ├── 2021_bcrast_pstahead.csv
│   ├── 2021_bcrast_pstahead.nc
│   ├── 2021_bcrast_pstahuman.csv
│   ├── 2021_bcrast_pstahuman.nc
│   ├── 2021_bcrast_pstalightning.csv
│   ├── 2021_bcrast_pstalightning.nc
│   ├── 2022_bcrast_biomass.nc
│   ├── 2022_bcrast_forestage.nc
│   ├── 2022_bcrast_meanheight.nc
│   ├── 2022_bcrast_stddevheight.nc
│   ├── 2024_bcrast_fueltypes.csv
│   ├── 2024_bcrast_fueltypes.nc
│   ├── 2024_bcrast_ownership.csv
│   ├── 2024_bcrast_ownership.nc
│   ├── 2026_bcrast_dnbr.nc
│   ├── 2026_bcrast_nbacyear.nc
│   └── 2026_bcrast_salvagelogging.nc
├── bcvect
│   ├── 2020_bcvect_extent.gpkg
│   ├── 2022_bcvect_firezones.gpkg
│   └── 2024_bcvect_watersheds.gpkg
├── BCFireCube_Codebook.pdf
└── README.md
```

Wildfire Data

Point and Polygon Data

This file contains data values describing wildfires associated with both a fire point and a fire polygon.

File Name2026_bcfire_bothdata.csv

Observations12,590

Dataset Production Year 2026

Dataset Coverage Years 1950-2025

Variables

firelabel, label, date, igndate, outdate, burnlength, area_sqm, datasource,
creationnotes, cause, fire_centre, fire_zone, firetype, responsetype,
elevation, avgelevation, slope, avgslope, landcover, avglancover, beccode,
avgbeccode, ndtcode, avgndtcode, fueltype, avgfueltype, population,
avgpopulation, dwellings, avgdwellings, forestage, avgforestage, biomass,
avgbiomass

Variable Coverage Years

firelabel, label, date, igndate, outdate, burnlength, area_sqm,	1950-2025
datasource, creationnotes, cause, firetype, elevation,	
avgelevation, slope, avgslope	
fire_centre, fire_zone	1989-2025
responsetype	2000-2025
population, avgpopulation, dwellings, avgdwellings	2002-2025
landcover, avglancover	2006-2024
beccode, avgbeccode, ndtcode, avgndtcode	2017-2025
fueltype, avgfueltype	2020-2025
forestage, avgforestage, biomass, avgbiomass	2021-2023

Variable Descriptions

firelabel	Official wildfire identifier. Formatted as YYYY-#####.
label	Modified wildfire identifier. Differentiates between wildfires within the same complex. Formatted as YYYY-#####-#.
date	Fire polygon recorded date. Formatted as YYYY-MM-DD.
igndate	Fire point recorded ignition date. Formatted as YYYY-MM-DD.
outdate	Fire point recorded fire out date. Formatted as YYYY-MM-DD.
burnlength	Difference in days between igndate and outdate.

area_sqm	Fire polygon area in square metres.
datasource	Fire polygon recorded source of data upload method.
creationnotes	Fire polygon recorded additional information. Clarifies data source or potential inaccuracy.
cause	Recorded fire cause. All entries are “Human,” “Lightning,” or “Unknown.”
fire_centre	Fire point recorded fire centre.
fire_zone	Fire point recorded fire zone.
firetype	Fire point recorded fire type. All entries are “Fire” or “Nuisance Fire.”
responsetype	Fire point recorded response type. All entries are “Full,” “Modified,” or “Monitor.”
elevation	Fire point extracted elevation in metres.
avelevation	Fire polygon extracted mean elevation in metres.
slope	Fire point extracted slope in degrees.
avgslope	Fire polygon extracted mean slope in degrees.
landcover	Fire point extracted land cover code. See documentation table for specific values.
avglandcover	Fire polygon extracted mode land cover code. See documentation table for specific values.
beccode	Fire point extracted biogeoclimatic zone and subzone code. See documentation table for specific values.
avgbeccode	Fire polygon extracted mode biogeoclimatic zone and subzone code. See documentation table for specific values.
ndtcode	Fire point extracted natural disturbance type code. See documentation table for specific values.
avgndtcode	Fire polygon extracted mode natural disturbance type code. See documentation table for specific values.
fueltype	Fire point extracted fuel type code. See documentation table for specific values.
avgfueltype	Fire polygon extracted mode fuel type code. See documentation table for specific values.
population	Fire point extracted population density in count per square kilometre. A uniform population distribution within census dissemination blocks is assumed.
avgpopulation	Fire polygon extracted mean population density in count per square kilometre. A uniform population distribution within census dissemination blocks is assumed.
dwellings	Fire point extracted dwelling density in count per square kilometre. A uniform dwelling distribution within census dissemination blocks is assumed.
avgdwellings	Fire polygon extracted mean dwelling density in count per square kilometre. A uniform dwelling distribution within census dissemination blocks is assumed.

forestage	Fire point extracted forest age in first growth year. A value of “1871” indicates that the forest began growing during or before 1871. A value of “0” indicates no forest cover.
avgforestage	Fire polygon extracted mode forest age in first growth year. A value of “1871” indicates that the forest began growing during or before 1871. A value of “0” indicates no forest cover.
biomass	Fire point extracted forest total biomass in tons per square kilometre.
avgbiomass	Fire polygon extracted mean forest total biomass in tons per square kilometre.

Licence Information

This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada. Adapted from Statistics Canada, 92-151-X and 92-163-X, 2006, 2011, 2016, and 2021. This does not constitute an endorsement by Statistics Canada of this product.

Citations

- BC Wildfire Service. BC Wildfire Perimeters - Historical. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bc-wildfire-fire-perimeters-historical> (2025).
- BC Wildfire Service. BC Wildfire Fire Incident Locations - Historical. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bc-wildfire-fire-incident-locations-historical> (2025).
- GeoBC. Digital Elevation Model for British Columbia - CDED - 1:250,000. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/digital-elevation-model-for-british-columbia-cded-1-250-000> (2014).
- Canada Centre for Remote Sensing. 2020 Land Cover of Canada. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/ee1580ab-a23d-4f86-a09b-79763677eb47> (2022).
- Canada Centre for Remote Sensing. 2015 Land Cover of Canada. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/4e615eae-b90c-420b-a000-2ca35896caf6> (2022).
- Canada Centre for Remote Sensing. 2010 Land Cover of Canada. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/c688b87f-e85f-4842-b0e1-a8f79ebf1133> (2022).
- Forest Analysis and Inventory Branch. BEC Map. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bec-map> (2021).
- Forest Analysis and Inventory Branch. Natural Disturbance Type Map. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/natural-disturbance-type-map> (2021).
- Canadian Forest Service. Canadian Forest Fire Danger Rating System (CFFDRS) Fire Behaviour Prediction (FBP) Fuel Types 2024, 30 M. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/4e66dd2f-5cd0-42fd-b82c-a430044b31de> (2024).

- Statistics Canada. Geographic Attribute File, 2006 Census. *Statistics Canada Catalogue* no. 92-151-X.
- Statistics Canada. Dissemination Block Boundary File, 2006 Census. *Statistics Canada Catalogue* no. 92-163-X.
- Statistics Canada. Geographic Attribute File, 2011 Census. *Statistics Canada Catalogue* no. 92-151-X.
- Statistics Canada. Dissemination Block Boundary File, 2011 Census. *Statistics Canada Catalogue* no. 92-163-X.
- Statistics Canada. Geographic Attribute File, 2016 Census. *Statistics Canada Catalogue* no. 92-151-X.
- Statistics Canada. Dissemination Block Boundary File, 2016 Census. *Statistics Canada Catalogue* no. 92-163-X.
- Statistics Canada. Geographic Attribute File, 2021 Census. *Statistics Canada Catalogue* no. 92-151-X.
- Statistics Canada. Dissemination Block Boundary File, 2021 Census. *Statistics Canada Catalogue* no. 92-163-X.
- Canadian Forest Service. Forest Age (2022). *Open Government Portal*. <https://open.canada.ca/data/en/dataset/2dac1be0-411b-421e-bdf5-142a77afdbf0> (2023).
- Canadian Forest Service. Forest Total Biomass (2022). *Open Government Portal*. <https://open.canada.ca/data/en/dataset/877566da-647c-4e92-9e4e-357698575d59> (2026).
- GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Fire Points

This file contains spatial data identifying wildfires associated with a fire point and describing the incident location of those wildfires.

File Name 2026_bcfire_firepoints.gpkg

Observations 181,934

Dataset Production Year 2026

Dataset Coverage Years 1950-2025

Variables
firelabel, label, geom

Variable Coverage Years
firelabel, label, geom 1950-2025

Variable Descriptions
firelabel Official wildfire identifier. Formatted as YYYY-#####.
label Modified wildfire identifier. Differentiates between wildfires within the same complex. Formatted as YYYY-#####-#.
geom Fire point spatial data.

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
BC Wildfire Service. BC Wildfire Fire Incident Locations - Historical. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bc-wildfire-fire-incident-locations-historical> (2025).
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Fire Polygons

This file contains spatial data identifying wildfires associated with a fire polygon and describing the fire perimeter of those wildfires.

File Name2026_bcfire_firepolygons.gpkg

Observations24,085

Dataset Production Year 2026

Dataset Coverage Years 1917-2025

Variables
firelabel, label, geom

Variable Coverage Years
firelabel, label, geom 1917-2025

Variable Descriptions
firelabel Official wildfire identifier. Formatted as YYYY-#####.
label Modified wildfire identifier. Differentiates between wildfires within the same complex. Formatted as YYYY-#####-#.
geom Fire polygon spatial data.

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
BC Wildfire Service. BC Wildfire Perimeters - Historical. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bc-wildfire-fire-perimeters-historical> (2025).

Point Data

This file contains data values describing wildfires associated with a fire point.

File Name2026_bcfire_pointdata.csv

Observations 181,934

Dataset Production Year 2026

Dataset Coverage Years 1950-2025

Variables
firelabel, label, igndate, outdate, burnlength, cause, fire_centre, fire_zone,
firetype, responsetype, elevation, slope, landcover, beccode, ndtcode,
fueltype, population, dwellings, forestage, biomass

Variable Coverage Years
firelabel, label, igndate, outdate, burnlength, cause, 1950-2025
firetype, elevation, slope
fire_centre, fire_zone 1989-2025
responsetype 2000-2025
population, dwellings 2002-2025
landcover 2006-2024
beccode, ndtcode 2017-2025
fueltype 2020-2025
forestage, biomass 2021-2023

Variable Descriptions
firelabel Official wildfire identifier. Formatted as YYYY-#####.
label Modified wildfire identifier. Differentiates between wildfires
within the same complex. Formatted as YYYY-#####-#.
igndate Fire point recorded ignition date. Formatted as YYYY-MM-DD.
outdate Fire point recorded fire out date. Formatted as YYYY-MM-DD.
burnlength Difference in days between igndate and outdate.
cause Recorded fire cause. All entries are “Human,” “Lightning,” or
“Unknown.”
fire_centre Fire point recorded fire centre.
fire_zone Fire point recorded fire zone.
firetype Fire point recorded fire type. All entries are “Fire” or “Nuisance
Fire.”
responsetype Fire point recorded response type. All entries are “Full,” “Mod-
ified,” or “Monitor.”
elevation Fire point extracted elevation in metres.
slope Fire point extracted slope in degrees.

landcover	Fire point extracted land cover code. See documentation table for specific values.
beccode	Fire point extracted biogeoclimatic zone and subzone code. See documentation table for specific values.
ndtcode	Fire point extracted natural disturbance type code. See documentation table for specific values.
fueltype	Fire point extracted fuel type code. See documentation table for specific values.
population	Fire point extracted population density in count per square kilometre. A uniform population distribution within census dissemination blocks is assumed.
dwellings	Fire point extracted dwelling density in count per square kilometre. A uniform dwelling distribution within census dissemination blocks is assumed.
forestage	Fire point extracted forest age in first growth year. A value of “1871” indicates that the forest began growing during or before 1871. A value of “0” indicates no forest cover.
biomass	Fire point extracted forest total biomass in tons per square kilometre.

Licence Information

This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada. Adapted from Statistics Canada, 92-151-X and 92-163-X, 2006, 2011, 2016, and 2021. This does not constitute an endorsement by Statistics Canada of this product.

Citations

BC Wildfire Service. BC Wildfire Fire Incident Locations - Historical. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bc-wildfire-fire-incident-locations-historical> (2025).

GeoBC. Digital Elevation Model for British Columbia - CDED - 1:250,000. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/digital-elevation-model-for-british-columbia-cded-1-250-000> (2014).

Canada Centre for Remote Sensing. 2020 Land Cover of Canada. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/ee1580ab-a23d-4f86-a09b-79763677eb47> (2022).

Canada Centre for Remote Sensing. 2015 Land Cover of Canada. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/4e615eae-b90c-420b-a000-2ca35896caf6> (2022).

Canada Centre for Remote Sensing. 2010 Land Cover of Canada. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/c688b87f-e85f-4842-b0e1-a8f79ebf1133> (2022).

Forest Analysis and Inventory Branch. BEC Map. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bec-map> (2021).

- Forest Analysis and Inventory Branch. Natural Disturbance Type Map. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/natural-disturbance-type-map> (2021).
- Canadian Forest Service. Canadian Forest Fire Danger Rating System (CFFDRS) Fire Behaviour Prediction (FBP) Fuel Types 2024, 30 M. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/4e66dd2f-5cd0-42fd-b82c-a430044b31de> (2024).
- Statistics Canada. Geographic Attribute File, 2006 Census. *Statistics Canada Catalogue* no. 92-151-X.
- Statistics Canada. Dissemination Block Boundary File, 2006 Census. *Statistics Canada Catalogue* no. 92-163-X.
- Statistics Canada. Geographic Attribute File, 2011 Census. *Statistics Canada Catalogue* no. 92-151-X.
- Statistics Canada. Dissemination Block Boundary File, 2011 Census. *Statistics Canada Catalogue* no. 92-163-X.
- Statistics Canada. Geographic Attribute File, 2016 Census. *Statistics Canada Catalogue* no. 92-151-X.
- Statistics Canada. Dissemination Block Boundary File, 2016 Census. *Statistics Canada Catalogue* no. 92-163-X.
- Statistics Canada. Geographic Attribute File, 2021 Census. *Statistics Canada Catalogue* no. 92-151-X.
- Statistics Canada. Dissemination Block Boundary File, 2021 Census. *Statistics Canada Catalogue* no. 92-163-X.
- Canadian Forest Service. Forest Age (2022). *Open Government Portal*. <https://open.canada.ca/data/en/dataset/2dac1be0-411b-421e-bdf5-142a77afdbf0> (2023).
- Canadian Forest Service. Forest Total Biomass (2022). *Open Government Portal*. <https://open.canada.ca/data/en/dataset/877566da-647c-4e92-9e4e-357698575d59> (2026).
- GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Polygon Data

This file contains data values describing wildfires associated with a fire polygon.

File Name 2026_bcfire_polygondata.csv

Observations 24,085

Dataset Production Year 2026

Dataset Coverage Years 1917-2025

Variables
firelabel, label, date, cause, area_sqm, datasource, creationnotes,
avgelevation, avgslope, avglandcover, avgbeccode, avgndtcode, avgfueltype,
avgpopulation, avgdwellings, avgforestage, avgbiomass

Variable Coverage Years
firelabel, label, date, cause, area_sqm, datasource, 1917-2025
creationnotes, avgelevation, avgslope
avgpopulation, avgdwellings 2002-2025
avglandcover 2006-2024
avgbeccode, avgndtcode 2017-2025
avgfueltype 2020-2025
avgforestage, avgbiomass 2021-2023

Variable Descriptions
firelabel Official wildfire identifier. Formatted as YYYY-#####.
label Modified wildfire identifier. Differentiates between wildfires
within the same complex. Formatted as YYYY-#####-#.
date Fire polygon recorded date. Formatted as YYYY-MM-DD.
cause Recorded fire cause. All entries are “Human,” “Lightning,” or
“Unknown.”
area_sqm Fire polygon area in square metres.
datasource Fire polygon recorded source of data upload method.
creationnotes Fire polygon recorded additional information. Clarifies data
source or potential inaccuracy.
avgelevation Fire polygon extracted mean elevation in metres.
avgslope Fire polygon extracted mean slope in degrees.
avglandcover Fire polygon extracted mode land cover code. [See documentation
table for specific values.](#)
avgbeccode Fire polygon extracted mode biogeoclimatic zone and subzone
code. [See documentation table for specific values.](#)

avgnndtcode	Fire polygon extracted mode natural disturbance type code. See documentation table for specific values .
avgfueltype	Fire polygon extracted mode fuel type code. See documentation table for specific values .
avgpopulation	Fire polygon extracted mean population density in count per square kilometre. A uniform population distribution within census dissemination blocks is assumed.
avgdwellings	Fire polygon extracted mean dwelling density in count per square kilometre. A uniform dwelling distribution within census dissemination blocks is assumed.
avgforestage	Fire polygon extracted mode forest age in first growth year. A value of “1871” indicates that the forest began growing during or before 1871. A value of “0” indicates no forest cover.
avgbiomass	Fire polygon extracted mean forest total biomass in tons per square kilometre.

Licence Information

This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada. Adapted from Statistics Canada, 92-151-X and 92-163-X, 2006, 2011, 2016, and 2021. This does not constitute an endorsement by Statistics Canada of this product. Contains “Access Only” information licensed with copyright permission from His Majesty the King in right of the Province of British Columbia.

Citations

- BC Wildfire Service. BC Wildfire Perimeters - Historical. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bc-wildfire-fire-perimeters-historical> (2025).
- GeoBC. Digital Elevation Model for British Columbia - CDED - 1:250,000. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/digital-elevation-model-for-british-columbia-cded-1-250-000> (2014).
- Canada Centre for Remote Sensing. 2020 Land Cover of Canada. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/ee1580ab-a23d-4f86-a09b-79763677eb47> (2022).
- Canada Centre for Remote Sensing. 2015 Land Cover of Canada. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/4e615eae-b90c-420b-adee-2ca35896caf6> (2022).
- Canada Centre for Remote Sensing. 2010 Land Cover of Canada. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/c688b87f-e85f-4842-b0e1-a8f79ebf1133> (2022).
- Forest Analysis and Inventory Branch. BEC Map. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bec-map> (2021).

- Forest Analysis and Inventory Branch. Natural Disturbance Type Map. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/natural-disturbance-type-map> (2021).
- Canadian Forest Service. Canadian Forest Fire Danger Rating System (CFFDRS) Fire Behaviour Prediction (FBP) Fuel Types 2024, 30 M. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/4e66dd2f-5cd0-42fd-b82c-a430044b31de> (2024).
- Statistics Canada. Geographic Attribute File, 2006 Census. *Statistics Canada Catalogue* no. 92-151-X.
- Statistics Canada. Dissemination Block Boundary File, 2006 Census. *Statistics Canada Catalogue* no. 92-163-X.
- Statistics Canada. Geographic Attribute File, 2011 Census. *Statistics Canada Catalogue* no. 92-151-X.
- Statistics Canada. Dissemination Block Boundary File, 2011 Census. *Statistics Canada Catalogue* no. 92-163-X.
- Statistics Canada. Geographic Attribute File, 2016 Census. *Statistics Canada Catalogue* no. 92-151-X.
- Statistics Canada. Dissemination Block Boundary File, 2016 Census. *Statistics Canada Catalogue* no. 92-163-X.
- Statistics Canada. Geographic Attribute File, 2021 Census. *Statistics Canada Catalogue* no. 92-151-X.
- Statistics Canada. Dissemination Block Boundary File, 2021 Census. *Statistics Canada Catalogue* no. 92-163-X.
- Canadian Forest Service. Forest Age (2022). *Open Government Portal*. <https://open.canada.ca/data/en/dataset/2dac1be0-411b-421e-bdf5-142a77afdbf0> (2023).
- Canadian Forest Service. Forest Total Biomass (2022). *Open Government Portal*. <https://open.canada.ca/data/en/dataset/877566da-647c-4e92-9e4e-357698575d59> (2026).
- GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Raster Data

Dwellings

These files contain raster data describing the mean dwelling density in count per square kilometre within census dissemination blocks for the 2006, 2011, 2016, and 2021 censuses.

File Names
2006_bcrast_dwellings.nc, 2011_bcrast_dwellings.nc,
2016_bcrast_dwellings.nc, 2021_bcrast_dwellings.nc

Dataset Production Years2006, 2011, 2016, 2021

Variable dwellings

Variable Description
dwellings Mean dwelling density in count per square kilometre. A uniform dwelling distribution within census dissemination blocks is assumed.

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada. Adapted from Statistics Canada, 92-151-X and 92-163-X, 2006, 2011, 2016, and 2021. This does not constitute an endorsement by Statistics Canada of this product.

Citations
Statistics Canada. Geographic Attribute File, 2006 Census. *Statistics Canada Catalogue* no. 92-151-X.
Statistics Canada. Dissemination Block Boundary File, 2006 Census. *Statistics Canada Catalogue* no. 92-163-X.
Statistics Canada. Geographic Attribute File, 2011 Census. *Statistics Canada Catalogue* no. 92-151-X.
Statistics Canada. Dissemination Block Boundary File, 2011 Census. *Statistics Canada Catalogue* no. 92-163-X.
Statistics Canada. Geographic Attribute File, 2016 Census. *Statistics Canada Catalogue* no. 92-151-X.
Statistics Canada. Dissemination Block Boundary File, 2016 Census. *Statistics Canada Catalogue* no. 92-163-X.
Statistics Canada. Geographic Attribute File, 2021 Census. *Statistics Canada Catalogue* no. 92-151-X.
Statistics Canada. Dissemination Block Boundary File, 2021 Census. *Statistics Canada Catalogue* no. 92-163-X.
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Land Cover

These files contain raster data describing the land cover classes of British Columbia in 2010, 2015, and 2020.

File Names
2010_bcrast_landcover.csv, 2010_bcrast_landcover.nc,
2015_bcrast_landcover.csv, 2015_bcrast_landcover.nc,
2020_bcrast_landcover.csv, 2020_bcrast_landcover.nc

Dataset Production Years2010, 2015, 2020

Variables
landcover, Land.Cover.Class

Variable Descriptions
landcover Land cover code. See documentation table for specific values.
Land.Cover.Class Land cover code longname description. See documentation table for specific values.

Documentation Table

landcover	Land.Cover.Class
1	Temperate or sub-polar needleleaf forest
2	Sub-polar taiga needleleaf forest
5	Temperate or sub-polar broadleaf deciduous forest
6	Mixed forest
8	Temperate or sub-polar shrubland
10	Temperate or sub-polar grassland
11	Sub-polar or polar shrubland-lichen-moss
12	Sub-polar or polar grassland-lichen-moss
14	Wetland
15	Cropland
16	Barren land
17	Urban and built-up
18	Water
19	Snow and ice

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
Canada Centre for Remote Sensing. 2020 Land Cover of Canada. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/ee1580ab-a23d-4f86-a09b-79763677eb47> (2022).

- Canada Centre for Remote Sensing. 2015 Land Cover of Canada. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/4e615eae-b90c-420b-a-dee-2ca35896caf6> (2022).
- Canada Centre for Remote Sensing. 2010 Land Cover of Canada. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/c688b87f-e85f-4842-b0e1-a8f79ebf1133> (2022).
- GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Population

These files contain raster data describing the mean population density in count per square kilometre within census dissemination blocks for the 2006, 2011, 2016, and 2021 censuses.

File Names
2006_bcrast_population.nc, 2011_bcrast_population.nc,
2016_bcrast_population.nc, 2021_bcrast_population.nc

Dataset Production Year2006, 2011, 2016, 2021

Variable population

Variable Description
population Mean population density in count per square kilometre. A uniform population distribution within census dissemination blocks is assumed.

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada. Adapted from Statistics Canada, 92-151-X and 92-163-X, 2006, 2011, 2016, and 2021. This does not constitute an endorsement by Statistics Canada of this product.

Citations
Statistics Canada. Geographic Attribute File, 2006 Census. *Statistics Canada Catalogue* no. 92-151-X.
Statistics Canada. Dissemination Block Boundary File, 2006 Census. *Statistics Canada Catalogue* no. 92-163-X.
Statistics Canada. Geographic Attribute File, 2011 Census. *Statistics Canada Catalogue* no. 92-151-X.
Statistics Canada. Dissemination Block Boundary File, 2011 Census. *Statistics Canada Catalogue* no. 92-163-X.
Statistics Canada. Geographic Attribute File, 2016 Census. *Statistics Canada Catalogue* no. 92-151-X.
Statistics Canada. Dissemination Block Boundary File, 2016 Census. *Statistics Canada Catalogue* no. 92-163-X.
Statistics Canada. Geographic Attribute File, 2021 Census. *Statistics Canada Catalogue* no. 92-151-X.
Statistics Canada. Dissemination Block Boundary File, 2021 Census. *Statistics Canada Catalogue* no. 92-163-X.
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Digital Elevation Model

This file contains raster data describing the topographic elevation of British Columbia.

File Name 2014_bcrast_dem.nc

Dataset Production Year 2014

Variable elevation

Variable Description
elevation Topographic elevation in metres.

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
GeoBC. Digital Elevation Model for British Columbia - CDED - 1:250,000. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/digital-elevation-model-for-british-columbia-cded-1-250-000> (2014).
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Slope

This file contains raster data describing the topographic slope of British Columbia.

File Name2014_bcrast_slope.nc

Dataset Production Year 2014

Variable slope

Variable Description
slope Topographic slope in degrees.

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
GeoBC. Digital Elevation Model for British Columbia - CDED - 1:250,000. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/digital-elevation-model-for-british-columbia-cded-1-250-000> (2014).
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Wildland-Urban Interface Buffer

This file contains raster data describing a two kilometre wildland-urban interface buffer around dense development in British Columbia.

File Name2020_bcrast_wui.nc

Dataset Production Year 2020

Variablewui

Variable Description
wui Binary variable describing the extent of a two kilometre wildland-urban interface buffer around dense development. A value of “1” indicates that the buffer is present.

Licence Information
This file is licensed under CC BY-NC-SA 4.0: <https://creativecommons.org/licenses/by-nc-sa/4.0/>. Contains information licensed under the Open Government Licence—Canada. Contains “Access Only” information licensed with copyright permission from His Majesty the King in right of the Province of British Columbia.

Citations
BC Wildfire Service. BC Wildfire WUI Human Interface Buffer. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bc-wildfire-wui-human-interface-buffer> (2020).
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Biogeoclimatic Zone and Subzone

This file contains raster data describing the biogeoclimatic zones and subzones of British Columbia.

File Names
2021_bcrast_bec.csv, 2021_bcrast_bec.nc

Dataset Production Year 2021

Variables
beccode, Z_SZ, ZONE, SUBZONE, ZONE_NAME, SBZNNM

Variable Descriptions
 beccode Biogeoclimatic zone and subzone code. See documentation table for specific values.
 Z_SZ Biogeoclimatic zone and subzone hyphenated label identifier. See documentation table for specific values.
 ZONE Biogeoclimatic zone label identifier. See documentation table for specific values.
 SUBZONE Biogeoclimatic subzone label identifier. See documentation table for specific values.
 ZONE_NAME Biogeoclimatic zone longname description. See documentation table for specific values.
 SBZNNM Biogeoclimatic subzone longname description. See documentation table for specific values.

Documentation Table						
beccode	Z_SZ	ZONE	SUBZONE	ZONE_NAME	SBZNNM	
10	PP-xh	PP	xh	Ponderosa Pine	Very Dry Hot	
20	CMA-un	CMA	un	Coastal Mountain-heather Alpine	Undifferentiated	
21	CMA-wh	CMA	wh	Coastal Mountain-heather Alpine	Wet Hypermaritime	
22	CMA-unp	CMA	unp	Coastal Mountain-heather Alpine	Undifferentiated and Parkland	
30	CDF-mm	CDF	mm	Coastal Douglas-fir	Moist Maritime	
40	IMA-un	IMA	un	Interior Mountain-heather Alpine	Undifferentiated	
41	IMA-unp	IMA	unp	Interior Mountain-heather Alpine	Undifferentiated and Parkland	
50	BG-xh	BG	xh	Bunchgrass	Very Dry Hot	
51	BG-xw	BG	xw	Bunchgrass	Very Dry Warm	
60	SBPS-dc	SBPS	dc	Sub-Boreal Pine – Spruce	Dry Cold	

Biogeoclimatic Zone and Subzone

beccode	Z_SZ	ZONE	SUBZONE	ZONE_NAME	SBZNM
61	SBPS-mc	SBPS	mc	Sub-Boreal Pine – Spruce	Moist Cold
62	SBPS-xc	SBPS	xc	Sub-Boreal Pine – Spruce	Very Dry Cold
63	SBPS-mk	SBPS	mk	Sub-Boreal Pine – Spruce	Moist Cool
70	SWB-uns	SWB	uns	Spruce – Willow Birch	Undifferentiated scrub
71	SWB-vk	SWB	vk	Spruce – Willow Birch	Very Wet Cool
72	SWB-mks	SWB	mks	Spruce – Willow Birch	Moist Cool Scrub
73	SWB-mk	SWB	mk	Spruce – Willow Birch	Moist Cool
74	SWB-un	SWB	un	Spruce – Willow Birch	Undifferentiated
75	SWB-vks	SWB	vks	Spruce – Willow Birch	Very Wet Cool Scrub
80	MH-mm	MH	mm	Mountain Hemlock	Moist Maritime
81	MH-wh	MH	wh	Mountain Hemlock	Wet Hypermaritime
82	MH-un	MH	un	Mountain Hemlock	Undifferentiated
83	MH-mmp	MH	mmp	Mountain Hemlock	Moist Maritime Parkland
84	MH-whp	MH	whp	Mountain Hemlock	Wet Hypermaritime Parkland
85	MH-unp	MH	unp	Mountain Hemlock	Undifferentiated parkland
90	BWBS-mk	BWBS	mk	Boreal White and Black Spruce	Moist Cool
91	BWBS-mw	BWBS	mw	Boreal White and Black Spruce	Moist Warm
92	BWBS-dk	BWBS	dk	Boreal White and Black Spruce	Dry Cool
93	BWBS-wk	BWBS	wk	Boreal White and Black Spruce	Wet Cool
94	BWBS-vk	BWBS	vk	Boreal White and Black Spruce	Very Wet Cool
100	MS-xv	MS	xv	Montane Spruce	Very Dry Very Cold
101	MS-dc	MS	dc	Montane Spruce	Dry Cold
102	MS-dk	MS	dk	Montane Spruce	Dry Cool
103	MS-dm	MS	dm	Montane Spruce	Dry Mild
104	MS-dv	MS	dv	Montane Spruce	Dry Very Cold

beccode	Z_SZ	ZONE	SUBZONE	ZONE_NAME	SBZNM
105	MS-xk	MS	xk	Montane Spruce	Very Dry Cool
106	MS-dw	MS	dw	Montane Spruce	Dry Warm
110	CWH-vh	CWH	vh	Coastal Western Hemlock	Very Wet Hypermaritime
111	CWH-ds	CWH	ds	Coastal Western Hemlock	Dry Submaritime
112	CWH-mm	CWH	mm	Coastal Western Hemlock	Moist Maritime
113	CWH-vm	CWH	vm	Coastal Western Hemlock	Very Wet Maritime
114	CWH-ms	CWH	ms	Coastal Western Hemlock	Moist Submaritime
115	CWH-wh	CWH	wh	Coastal Western Hemlock	Wet Hypermaritime
116	CWH-wm	CWH	wm	Coastal Western Hemlock	Wet Maritime
117	CWH-ws	CWH	ws	Coastal Western Hemlock	Wet Submaritime
118	CWH-xm	CWH	xm	Coastal Western Hemlock	Very Dry Maritime
119	CWH-dm	CWH	dm	Coastal Western Hemlock	Dry Maritime
120	SBS-mc	SBS	mc	Sub-Boreal Spruce	Moist Cold
121	SBS-dw	SBS	dw	Sub-Boreal Spruce	Dry Warm
122	SBS-dh	SBS	dh	Sub-Boreal Spruce	Dry Hot
123	SBS-mm	SBS	mm	Sub-Boreal Spruce	Moist Mild
124	SBS-mw	SBS	mw	Sub-Boreal Spruce	Moist Warm
125	SBS-un	SBS	un	Sub-Boreal Spruce	Undifferentiated
126	SBS-wk	SBS	wk	Sub-Boreal Spruce	Wet Cool
127	SBS-mh	SBS	mh	Sub-Boreal Spruce	Moist Hot
128	SBS-mk	SBS	mk	Sub-Boreal Spruce	Moist Cool
129	SBS-vk	SBS	vk	Sub-Boreal Spruce	Very Wet Cool
130	SBS-dk	SBS	dk	Sub-Boreal Spruce	Dry Cool
140	IDF-dk	IDF	dk	Interior Douglas-fir	Dry Cool
141	IDF-dm	IDF	dm	Interior Douglas-fir	Dry Mild
142	IDF-dw	IDF	dw	Interior Douglas-fir	Dry Warm
143	IDF-xh	IDF	xh	Interior Douglas-fir	Very Dry Hot
144	IDF-xm	IDF	xm	Interior Douglas-fir	Very Dry Mild
145	IDF-xw	IDF	xw	Interior Douglas-fir	Very Dry Warm
146	IDF-mw	IDF	mw	Interior Douglas-fir	Moist Warm
147	IDF-ww	IDF	ww	Interior Douglas-fir	Wet Warm
148	IDF-xk	IDF	xk	Interior Douglas-fir	Very Dry Cool
149	IDF-dc	IDF	dc	Interior Douglas-fir	Dry Cold
150	IDF-xc	IDF	xc	Interior Douglas-fir	Very Dry Cold

Biogeoclimatic Zone and Subzone

beccode	Z_SZ	ZONE	SUBZONE	ZONE_NAME	SBZNM
151	IDF-xx	IDF	xx	Interior Douglas-fir	Very Dry Very Hot
152	IDF-dh	IDF	dh	Interior Douglas-fir	Dry Hot
160	ICH-wk	ICH	wk	Interior Cedar – Hemlock	Wet Cool
161	ICH-mk	ICH	mk	Interior Cedar – Hemlock	Moist Cool
162	ICH-mm	ICH	mm	Interior Cedar – Hemlock	Moist Mild
163	ICH-mw	ICH	mw	Interior Cedar – Hemlock	Moist Warm
164	ICH-vc	ICH	vc	Interior Cedar – Hemlock	Very Wet Cold
165	ICH-vk	ICH	vk	Interior Cedar – Hemlock	Very Wet Cool
166	ICH-dk	ICH	dk	Interior Cedar – Hemlock	Dry Cool
167	ICH-mc	ICH	mc	Interior Cedar – Hemlock	Moist Cold
168	ICH-xw	ICH	xw	Interior Cedar – Hemlock	Very Dry Warm
169	ICH-wc	ICH	wc	Interior Cedar – Hemlock	Wet Cold
170	ICH-dw	ICH	dw	Interior Cedar – Hemlock	Dry Warm
171	ICH-dm	ICH	dm	Interior Cedar – Hemlock	Dry Mild
172	ICH-xm	ICH	xm	Interior Cedar – Hemlock	Very Dry Mild
180	ESSF-dc	ESSF	dc	Engelmann Spruce – Subalpine Fir	Dry Cold
181	ESSF-dvw	ESSF	dvw	Engelmann Spruce – Subalpine Fir	Dry Very Cold Woodland
182	ESSF-mmw	ESSF	mmw	Engelmann Spruce – Subalpine Fir	Moist Mild Woodland
183	ESSF-mw	ESSF	mw	Engelmann Spruce – Subalpine Fir	Moist Warm
184	ESSF-mww	ESSF	mww	Engelmann Spruce – Subalpine Fir	Moist Warm Woodland
185	ESSF-mh	ESSF	mh	Engelmann Spruce – Subalpine Fir	Moist Hot
186	ESSF-mkp	ESSF	mkp	Engelmann Spruce – Subalpine Fir	Moist Cool Parkland

beccode	Z_SZ	ZONE	SUBZONE	ZONE_NAME	SBZNM
187	ESSF-xv	ESSF	xv	Engelmann Spruce – Subalpine Fir	Very Dry Very Cold
188	ESSF-mk	ESSF	mk	Engelmann Spruce – Subalpine Fir	Moist Cool
189	ESSF-mv	ESSF	mv	Engelmann Spruce – Subalpine Fir	Moist Very Cold
190	ESSF-mm	ESSF	mm	Engelmann Spruce – Subalpine Fir	Moist Mild
191	ESSF-mwp	ESSF	mwp	Engelmann Spruce – Subalpine Fir	Moist Warm Parkland
192	ESSF-un	ESSF	un	Engelmann Spruce – Subalpine Fir	Undifferentiated
193	ESSF-vcp	ESSF	vcp	Engelmann Spruce – Subalpine Fir	Very Wet Cold Parkland
194	ESSF-vc	ESSF	vc	Engelmann Spruce – Subalpine Fir	Very Wet Cold
195	ESSF-wk	ESSF	wk	Engelmann Spruce – Subalpine Fir	Wet Cool
196	ESSF-wmp	ESSF	wmp	Engelmann Spruce – Subalpine Fir	Wet Mild Parkland
197	ESSF-wc	ESSF	wc	Engelmann Spruce – Subalpine Fir	Wet Cold
198	ESSF-xcp	ESSF	xcp	Engelmann Spruce – Subalpine Fir	Very Dry Cold Parkland
199	ESSF-mcp	ESSF	mcp	Engelmann Spruce – Subalpine Fir	Moist Cold Parkland
200	ESSF-wv	ESSF	wv	Engelmann Spruce – Subalpine Fir	Wet Very Cold
201	ESSF-dkp	ESSF	dkp	Engelmann Spruce – Subalpine Fir	Dry Cool Parkland
202	ESSF-mc	ESSF	mc	Engelmann Spruce – Subalpine Fir	Moist Cold
203	ESSF-dvp	ESSF	dvp	Engelmann Spruce – Subalpine Fir	Dry Very Cold Parkland
204	ESSF-wvp	ESSF	wvp	Engelmann Spruce – Subalpine Fir	Wet Very Cold Parkland
205	ESSF-dk	ESSF	dk	Engelmann Spruce – Subalpine Fir	Dry Cool
206	ESSF-dkw	ESSF	dkw	Engelmann Spruce – Subalpine Fir	Dry Cool Woodland
207	ESSF-dcp	ESSF	dcp	Engelmann Spruce – Subalpine Fir	Dry Cold Parkland
208	ESSF-dcw	ESSF	dcw	Engelmann Spruce – Subalpine Fir	Dry Cold Woodland

beccode	Z_SZ	ZONE	SUBZONE	ZONE_NAME	SBZNM
209	ESSF-mmp	ESSF	mmp	Engelmann Spruce – Subalpine Fir	Moist Mild Parkland
210	ESSF-mvp	ESSF	mvp	Engelmann Spruce – Subalpine Fir	Moist Very Cold Parkland
211	ESSF-wcp	ESSF	wcp	Engelmann Spruce – Subalpine Fir	Wet Cold Parkland
212	ESSF-wcw	ESSF	wcw	Engelmann Spruce – Subalpine Fir	Wet Cold Woodland
213	ESSF-wmw	ESSF	wmw	Engelmann Spruce – Subalpine Fir	Wet Mild Woodland
214	ESSF-xvp	ESSF	xvp	Engelmann Spruce – Subalpine Fir	Very Dry Very Cold Parkland
215	ESSF-unp	ESSF	unp	Engelmann Spruce – Subalpine Fir	Undifferentiated parkland
216	ESSF-dv	ESSF	dv	Engelmann Spruce – Subalpine Fir	Dry Very Cold
217	ESSF-xcw	ESSF	xcw	Engelmann Spruce – Subalpine Fir	Very Dry Cold Woodland
218	ESSF-vcw	ESSF	vcw	Engelmann Spruce – Subalpine Fir	Very Wet Cold Woodland
219	ESSF-xc	ESSF	xc	Engelmann Spruce – Subalpine Fir	Very Dry Cold
220	ESSF-xvw	ESSF	xvw	Engelmann Spruce – Subalpine Fir	Very Dry Very Cold Woodland
221	ESSF-mcw	ESSF	mcw	Engelmann Spruce – Subalpine Fir	Moist Cold Woodland
222	ESSF-wh	ESSF	wh	Engelmann Spruce – Subalpine Fir	Wet Hot
223	ESSF-mkw	ESSF	mkw	Engelmann Spruce – Subalpine Fir	Moist Cool Woodland
224	ESSF-wm	ESSF	wm	Engelmann Spruce – Subalpine Fir	Wet Mild
225	ESSF-dh	ESSF	dh	Engelmann Spruce – Subalpine Fir	Dry Hot
226	ESSF-wvw	ESSF	wvw	Engelmann Spruce – Subalpine Fir	Wet Very Cold Woodland
230	BAFA-un	BAFA	un	Boreal Altai Fescue Alpine	Undifferentiated
231	BAFA-unp	BAFA	unp	Boreal Altai Fescue Alpine	Undifferentiated and Parkland

Licence Information

This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations

- Forest Analysis and Inventory Branch. BEC Map. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bec-map> (2021).
- GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Natural Disturbance Type

This file contains raster data describing the natural disturbance type classes of British Columbia.

File Names
2021_bcrast_ndt.csv, 2021_bcrast_ndt.nc

Dataset Production Year 2021

Variables
ndtcode, NTRL_DSTRD, NTRL_DSTRC

Variable Descriptions
ndtcode Natural disturbance type class code. See documentation table for specific values.
NTRL_DSTRD Natural disturbance type class index. See documentation table for specific values.
NTRL_DSTRC Natural disturbance type class description. See documentation table for specific values.

Documentation Table

ndtcode	NTRL_DSTRD	NTRL_DSTRC
1	NDT1	Ecosystems with rare stand-initiating events
2	NDT2	Ecosystems with infrequent stand-initiating events
3	NDT3	Ecosystems with frequent stand-initiating events
4	NDT4	Ecosystems with frequent stand-maintaining fires
5	NDT5	Alpine Tundra and Subalpine Parkland ecosystems

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
Forest Analysis and Inventory Branch. Natural Disturbance Type Map. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/natural-disturbance-type-map> (2021).
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

PSTA Fire Density

This file contains raster data describing the Provincial Strategic Threat Analysis modelled number of nearby historical fire incidents since 1950. Fires that are greater than four hectares in area are treated as one event. Fires that are greater than five hundred hectares are treated as five events.

File Names
2021_bcrast_pstafire.csv, 2021_bcrast_pstafire.nc

Dataset Production Year 2021

Variables
fire_density, FR_STRT_DY

Variable Descriptions
fire_density Code representing range of PSTA modelled number of fire starts since 1950. See documentation table for specific values.
FR_STRT_DY Range of PSTA modelled number of fire starts since 1950. See documentation table for specific values.

Documentation Table

fire_density	FR_STRT_DY
-1	Water
0	No Fires
1	1 - 5
2	5.1 – 10
3	10.1 – 17
4	17.1 – 24
5	24.1 – 33
6	33.1 – 45
7	45.1 – 60
8	60.1 – 82
9	82.1 – 116
10	>116

Licence Information
This file is licensed under CC BY-NC-SA 4.0: <https://creativecommons.org/licenses/by-nc-sa/4.0/>. Contains information licensed under the Open Government Licence—Canada. Contains “Access Only” information licensed with copyright permission from His Majesty the King in right of the Province of British Columbia.

Citations
BC Wildfire Service. BC Wildfire PSTA Fire Density. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bc-wildfire-psta-fire-density> (2021).
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data*

Catalogue <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

PSTA Head Fire Intensity

This file contains raster data describing the Provincial Strategic Threat Analysis modelled head fire intensity thresholds derived from 90th percentile fire weather index values in kilowatts per metre.

File Names
2021_bcrast_pstahead.csv, 2021_bcrast_pstahead.nc

Dataset Production Year 2021

Variables
hf_intensity, Head.Fire.Intensity.Class.Range

Variable Descriptions
hf_intensity Code representing range of PSTA modelled head fire intensity thresholds derived from 90th percentile fire weather index values in kilowatts per metre. See documentation table for specific values.
Head.Fire.Intensity.Class.Range Range of PSTA modelled head fire intensity thresholds derived from 90th percentile fire weather index values in kilowatts per metre. See documentation table for specific values.

Documentation Table	
hf_intensity	Head.Fire.Intensity.Class.Range
-3	No Data (Private Land)
-2	No Data (Private Managed Forest Land)
-1	Water
0	Non-Fuel
1	>0 to 1000 kW/m
2	>1000 to 2000 kW/m
3	>2000 to 4000 kW/m
4	>4000 to 6000 kW/m
5	>6000 to 10000 kW/m
6	>10000 to 18000 kW/m
7	>18000 to 30000 kW/m
8	>30000 to 60000 kW/m
9	>60000 to 100000 kW/m
10	>100000 kW/m

Licence Information
This file is licensed under CC BY-NC-SA 4.0: <https://creativecommons.org/licenses/by-nc-sa/4.0/>. Contains information licensed under the Open Government Licence—Canada. Contains “Access Only” information licensed with copyright

permission from His Majesty the King in right of the Province of British Columbia.

Citations

- BC Wildfire Service. BC Wildfire PSTA Head Fire Intensity. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bc-wildfire-psta-head-fire-intensity> (2021).
- GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

PSTA Human Fire Density

This file contains raster data describing the Provincial Strategic Threat Analysis modelled number of nearby human-caused historical fire incidents since 1950. Fires that are greater than four hectares in area are treated as one event. Fires that are greater than five hundred hectares are treated as five events.

File Names
2021_bcrast_pstahuman.csv, 2021_bcrast_pstahuman.nc

Dataset Production Year 2021

Variables
human_density, FR_STRT_DY

Variable Descriptions
human_density Code representing range of PSTA modelled number of human-caused fire starts since 1950. See documentation table for specific values.
FR_STRT_DY Range of PSTA modelled number of human-caused fire starts since 1950. See documentation table for specific values.

Documentation Table

human_density	FR_STRT_DY
-1	Water
0	No Fires
1	1 - 5
2	5.1 – 10
3	10.1 – 17
4	17.1 – 24
5	24.1 – 33
6	33.1 – 45
7	45.1 – 60
8	60.1 – 82
9	82.1 – 116
10	>116

Licence Information
This file is licensed under CC BY-NC-SA 4.0: <https://creativecommons.org/licenses/by-nc-sa/4.0/>. Contains information licensed under the Open Government Licence—Canada. Contains “Access Only” information licensed with copyright permission from His Majesty the King in right of the Province of British Columbia.

Citations
BC Wildfire Service. BC Wildfire PSTA Human Fire Density. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bc-wildfire-psta-human-fire-density> (2021).

GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

PSTA Lightning Fire Density

This file contains raster data describing the Provincial Strategic Threat Analysis modelled number of nearby lightning-caused historical fire incidents since 1950. Fires that are greater than four hectares in area are treated as one event. Fires that are greater than five hundred hectares are treated as five events.

File Names
2021_bcrast_pstalightning.csv, 2021_bcrast_pstalightning.nc

Dataset Production Year 2021

Variables
lightning_density, FR_STRT_DY

Variable Descriptions
lightning_density Code representing range of PSTA modelled number of human-caused fire starts since 1950. See documentation table for specific values.
FR_STRT_DY Range of PSTA modelled number of human-caused fire starts since 1950. See documentation table for specific values.

Documentation Table	
lightning_density	FR_STRT_DY
-1	Water
0	No Fires
1	1 - 5
2	5.1 – 10
3	10.1 – 17
4	17.1 – 24
5	24.1 – 33
6	33.1 – 45
7	45.1 – 60
8	60.1 – 82
9	82.1 – 116
10	>116

Licence Information
This file is licensed under CC BY-NC-SA 4.0: <https://creativecommons.org/licenses/by-nc-sa/4.0/>. Contains information licensed under the Open Government Licence—Canada. Contains “Access Only” information licensed with copyright permission from His Majesty the King in right of the Province of British Columbia.

Citations

- BC Wildfire Service. BC Wildfire PSTA Lightning Fire Density. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/bc-wildfire-psta-lightning-fire-density> (2021).
- GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Forest Total Biomass

This file contains raster data describing the forest total biomass in tons per square kilometre.

File Name 2022_bcrast_biomass.nc

Dataset Production Year 2022

Variable biomass

Variable Description
biomass Forest total biomass in tons per square kilometre.

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
Canadian Forest Service. Forest Total Biomass (2022). *Open Government Portal*.
<https://open.canada.ca/data/en/dataset/877566da-647c-4e92-9e4e-357698575d59> (2026).
Matasci, G. *et al.* Three decades of forest structural dynamics over Canada’s forested ecosystems using Landsat time-series and lidar plots. *Remote Sens Environ* **216** (2018). <https://doi.org/10.1016/j.rse.2018.07.024>
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Forest Age

This file contains raster data describing forest age in first growth year.

File Name2022_bcrast_forestage.nc

Dataset Production Year 2022

Variable `forestage`

Variable Description
 `forestage` Forest age in first growth year. A value of “1871” indicates that the forest began growing during or before 1871. A value of “0” indicates no forest cover.

Licence Information
 This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
 Canadian Forest Service. Forest Age (2022). *Open Government Portal*. <https://open.canada.ca/data/en/dataset/2dac1be0-411b-421e-bdf5-142a77afdbf0> (2023).
 Maltman, J. C. *et al.* Estimating and mapping forest age across Canada’s forested ecosystems. *Remote Sens Environ* **290** (2023). <https://doi.org/10.1016/j.rse.2023.113529>
 GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Forest Mean Height

This file contains raster data describing forest mean height in metres.

File Name2022_bcrast_meanheight.nc

Dataset Production Year 2022

Variableheight

Variable Description
height Forest mean height in metres.

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
Canadian Forest Service. Forest Elevation Mean (2022). *Open Government Portal*.
<https://open.canada.ca/data/en/dataset/7e1b6612-a3b2-400c-8318-63d93ab2f1b7> (2023).
Matasci, G. *et al.* Three decades of forest structural dynamics over Canada's forested ecosystems using Landsat time-series and lidar plots. *Remote Sens Environ* **216** (2018). <https://doi.org/10.1016/j.rse.2018.07.024>
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Forest Height Standard Deviation

This file contains raster data describing forest height standard deviation in metres.

File Name 2022_bcrast_stddevheight.nc

Dataset Production Year 2022

Variable height

Variable Description
height Forest height standard deviation in metres.

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
Canadian Forest Service. Forest Elevation (Ht) Standard Deviation (2022). *Open Government Portal*. <https://open.canada.ca/data/en/dataset/1803f056-48d6-4bc2-9b14-045428d320db> (2026).
Matasci, G. *et al.* Three decades of forest structural dynamics over Canada's forested ecosystems using Landsat time-series and lidar plots. *Remote Sens Environ* **216** (2018). <https://doi.org/10.1016/j.rse.2018.07.024>
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Fuel Types

This file contains raster data describing the fuel types of British Columbia.

File Names
2024_bcrast_fueltypes.csv, 2024_bcrast_fueltypes.nc

Dataset Production Year 2024

Variables
fueltype, fuelname, fuellongname

Variable Descriptions
fueltype Fuel type code. See documentation table for specific values.
fuelname Fuel type abbreviated name. See documentation table for specific values.
fuellongname Fuel type longname. See documentation table for specific values.

Documentation Table		
fueltype	fuelname	fuellongname
1	C1	C-1 Spruce-Lichen Woodland
2	C2	C-2 Boreal Spruce
3	C3	C-3 Mature Jack or Lodgepole Pine
4	C4	C-4 Immature Jack or Lodgepole Pine
5	C5	C-5 Red and White Pine
7	C7	C-7 Ponderosa Pine / Douglas Fir
11	D1	D-1 Leafless Aspen
13	D1/2	D-1/D-2 Aspen
31	O1	O-1a Matted Grass
101	Non-fuel	Non-fuel
102	Water	Water
105	Vegetated Non-fuel	Vegetated Non-fuel
415	M1 C15	M-1 Boreal Mixedwood - Leafless (15% Conifer)
625	M1/2 C25	M-1/M-2 Boreal Mixedwood (25% Conifer)
650	M1/2 C50	M-1/M-2 Boreal Mixedwood (50% Conifer)
675	M1/2 C75	M-1/M-2 Boreal Mixedwood (75% Conifer)

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
Canadian Forest Service. Canadian Forest Fire Danger Rating System (CFFDRS) Fire Behaviour Prediction (FBP) Fuel Types 2024, 30 M. *Open Government Portal*. <https://open.canada.ca/data/en/dataset/4e66dd2f-5cd0-42fd-b82c-a430044b31de> (2024).

GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Forest Cover Ownership

This file contains raster data describing the forest cover ownership of British Columbia.

File Names
2024_bcrast_ownership.csv, 2024_bcrast_ownership.nc

Dataset Production Year 2024

Variables
ownership, Ownership.Description

Variable Descriptions
ownership Forest cover ownership code. See documentation table for specific values.
Ownership.Description Description of forest cover ownership. See documentation table for specific values.

Documentation Table	
ownership	Ownership.Description
40	Private
41	Private - Land Claim Settlement Area
50	Federal - Federal Reserve
51	Federal - National Park
52	Federal - Indian Reserve
53	Federal - Military Reserve
54	Federal - Dominion government Block/Federal Parcels
60	Crown - Conservancy Area, Ecological Reserve, Protected Area, Provincial Park
61	Crown - UREP (Use, Recreation and Enjoyment of the Public Reserve)
62	Crown - Forest Management Unit
63	Crown - Biodiversity, Mining, Tourism Area
64	Crown - Special Forest Management Area
65	Crown - Wildlife Management Area
66	Crown - Recreation Area
67	Crown - Heritage Site
68	Crown - Forest Recreation Reserves
69	Crown - Misc. Reserves
70	Crown Tenure - Timber Licence
72	Crown Tenure - Tree Farm Licence, Schedule B
74	Crown - Timber Alienated in Watershed
75	Crown Tenure - Christmas Tree Licence
77	Crown Tenure - Woodlot Licence, Schedule B

ownership	Ownership.Description
78	Crown Tenure - First Nations Woodland Licence
79	Crown Tenure - Community Forest Agreement, Schedule B
80	Crown - Municipal Parcels
81	Crown - Local/Regional Park
91	Unknown Ownership/Exceptions
99	Crown Lease - Misc. lease

Licence Information

This file is licensed under CC BY-NC-SA 4.0: <https://creativecommons.org/licenses/by-nc-sa/4.0/>. Contains information licensed under the Open Government Licence—Canada. Contains “Access Only” information licensed with copyright permission from His Majesty the King in right of the Province of British Columbia.

Citations

Forest Analysis and Inventory Branch. Generalized Forest Cover Ownership. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/generalized-forest-cover-ownership> (2024).

GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Differenced Normalized Burn Ratio

This file contains raster data describing the differenced Normalized Burn Ratio of burned areas throughout British Columbia.

File Name2026_bcrast_dnbr.nc

Dataset Production Year 2026

Variable dnbr

Variable Description
dnbr Differenced Normalized Burn Ratio value. A value of “-9999” corresponds to an unburned area.

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
Canadian Forest Service. Canada Landsat Burned Severity (CanLaBS v2): a Canada-wide Landsat-based 30m resolution product of burned severity since 1985. *Open Government Portal*. <https://doi.org/10.23687/2af751e7-79f9-4da8-9b45-14688818dca3> (2026).
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

NBAC Burn Year

This file contains raster data describing the burn year of different fires in the National Burn Area Composite.

File Name 2026_bcrast_nbacyear.nc

Dataset Production Year 2026

Variable burnyear

Variable Description
burnyear National Burn Area Composite burn year. A value of “0” corresponds to an unburned area.

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
Canadian Forest Service. Canada Landsat Burned Severity (CanLaBS v2): a Canada-wide Landsat-based 30m resolution product of burned severity since 1985. *Open Government Portal*. <https://doi.org/10.23687/2af751e7-79f9-4da8-9b45-14688818dca3> (2026).
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Salvage Logging

This file contains raster data describing whether salvage logging took place for locations in the National Burn Area Composite.

File Name2026_bcrast_salvagelogging.nc

Dataset Production Year 2026

Variable salvagelog

Variable Description
 salvagelog Binary code representing the presence of salvage logging in a burned area. A value of “1” indicates that salvage logging was present.

Licence Information
 This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
 Canadian Forest Service. Canada Landsat Burned Severity (CanLaBS v2): a Canada-wide Landsat-based 30m resolution product of burned severity since 1985. *Open Government Portal*. <https://doi.org/10.23687/2af751e7-79f9-4da8-9b45-14688818dca3> (2026).
 GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Vector Data

Terrestrial Extent

This file contains multipolygon vector data describing the terrestrial extent of British Columbia.

File Name 2020_bcvect_extent.gpkg

Observations 28,004

Dataset Production Year 2020

Variable geom

Variable Descriptions
geom Terrestrial extent spatial data.

Licence Information
This file is licensed under CC BY 4.0: <https://creativecommons.org/licenses/by/4.0/>. Contains information licensed under the Open Government Licence—Canada.

Citations
GeoBC. Province of British Columbia - Boundary Terrestrial. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/province-of-british-columbia-boundary-terrestrial> (2020).

Fire Centres and Zones

This file contains multipolygon vector data describing the administrative fire zones and centres of the British Columbia Wildfire Service.

File Name 2022_bcvect_firezones.gpkg

Observations32

Dataset Production Year 2022

Variables
fire_centre, fire_zone, geom

Variable Descriptions
fire_centre Name of fire centre.
fire_zone Name of fire zone.
geom Fire zone spatial data.

Licence Information
This file is licensed under CC BY-NC-SA 4.0: <https://creativecommons.org/licenses/by-nc-sa/4.0/>. Contains information licensed under the Open Government Licence—Canada. Contains “Access Only” information licensed with copyright permission from His Majesty the King in right of the Province of British Columbia.

Citations
BC Wildfire Service. BC Wildfire Fire Zones. *British Columbia Data Catalogue* <https://catalogue.data.gov.bc.ca/dataset/217bea05-4a6f-4ef8-8eb8-9b352e90828f> (2022).

Watershed Groups

This file contains multipolygon vector data describing the watershed groups of British Columbia.

File Name	2024_bcvect_watersheds.gpkg						
Observations	246						
Dataset Production Year	2024						
Variables	watershed_code, watershed_name, geom						
Variable Descriptions	<table><tr><td>watershed_code</td><td>Watershed group code in the format ####.</td></tr><tr><td>watershed_name</td><td>Name of watershed group.</td></tr><tr><td>geom</td><td>Watershed group spatial data.</td></tr></table>	watershed_code	Watershed group code in the format ####.	watershed_name	Name of watershed group.	geom	Watershed group spatial data.
watershed_code	Watershed group code in the format ####.						
watershed_name	Name of watershed group.						
geom	Watershed group spatial data.						
Licence Information	This file is licensed under CC BY 4.0: https://creativecommons.org/licenses/by/4.0/ . Contains information licensed under the Open Government Licence—Canada.						
Citations	GeoBC. Freshwater Atlas Watersheds. <i>British Columbia Data Catalogue</i> https://catalogue.data.gov.bc.ca/dataset/freshwater-atlas-watersheds (2008).						